

Model Based Clustering using MClust

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Introduction

We shall apply model based clustering on a dataset of bank note measurements

Load the packages and data. The data is built into the `mclust` package.

```
# install.packages(c("mclust", "patchwork", "ggridges"))

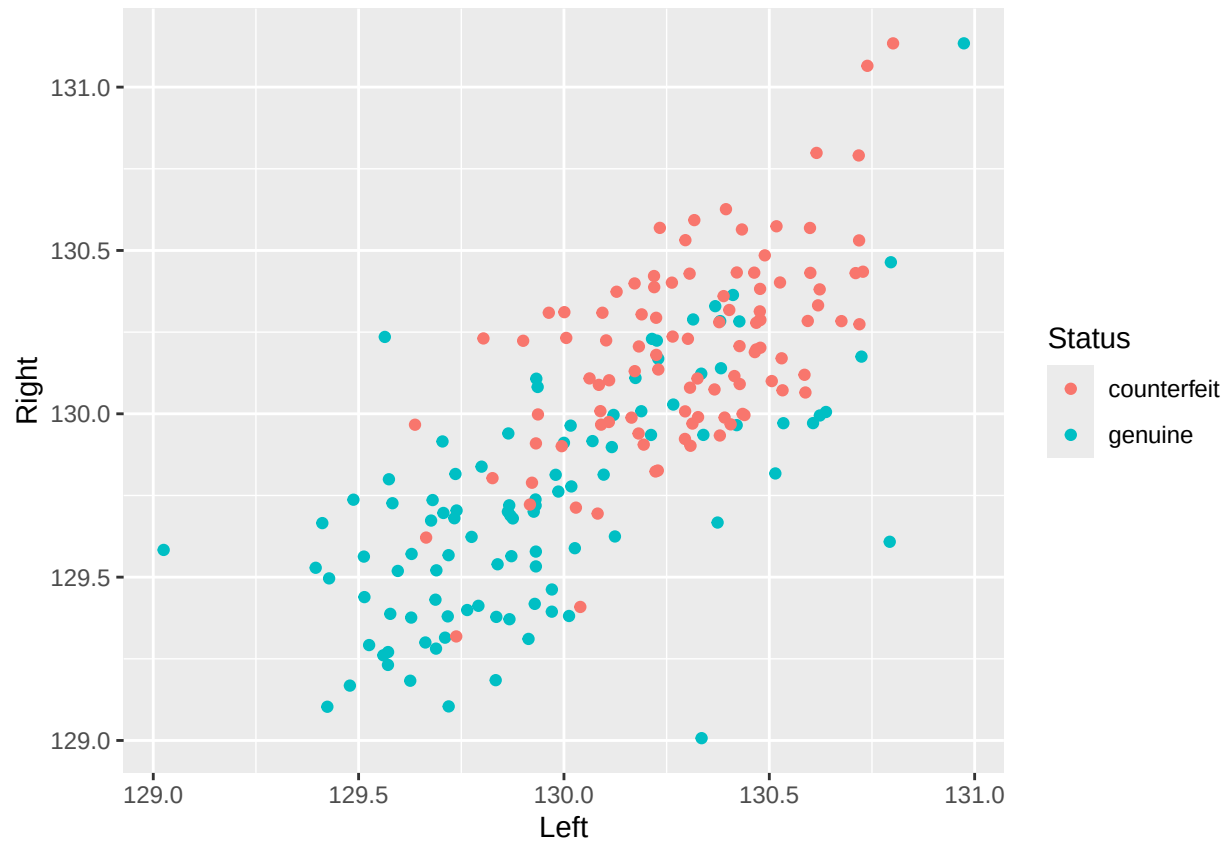
# load packages
library(mclust)
library(tidyverse)
library(patchwork)
library(ggridges)

# load data
df <- as.tibble(banknote)
```

Data Exploration

```
# help file to understand dataset
?banknote

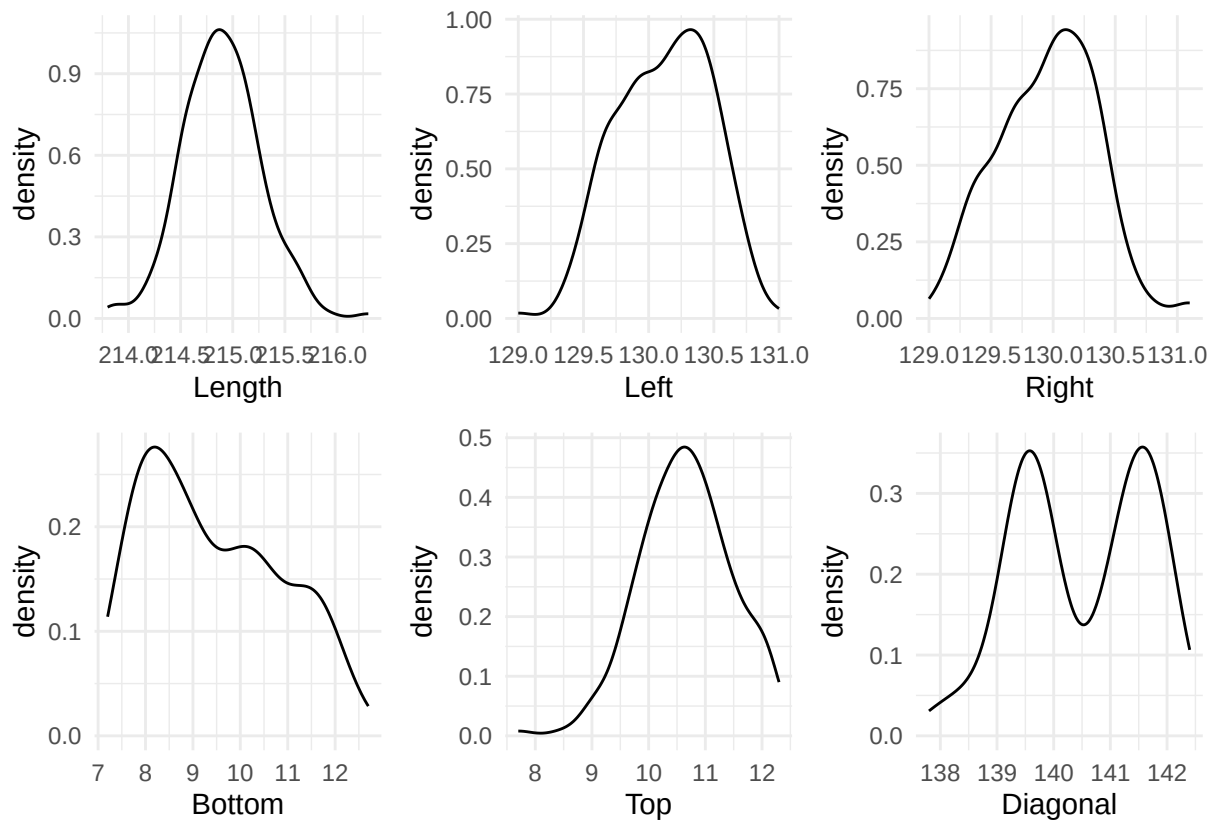
# scatter plot
df |>
  ggplot(aes(x = Left, y = Right, colour = Status)) +
  geom_point(position = position_jitter())
```



```
# The classes are not easy to distinguish, there is considerable overlap between
# the two classes
```

```
# remove Status column from the data set
df <- df |> select(-Status)

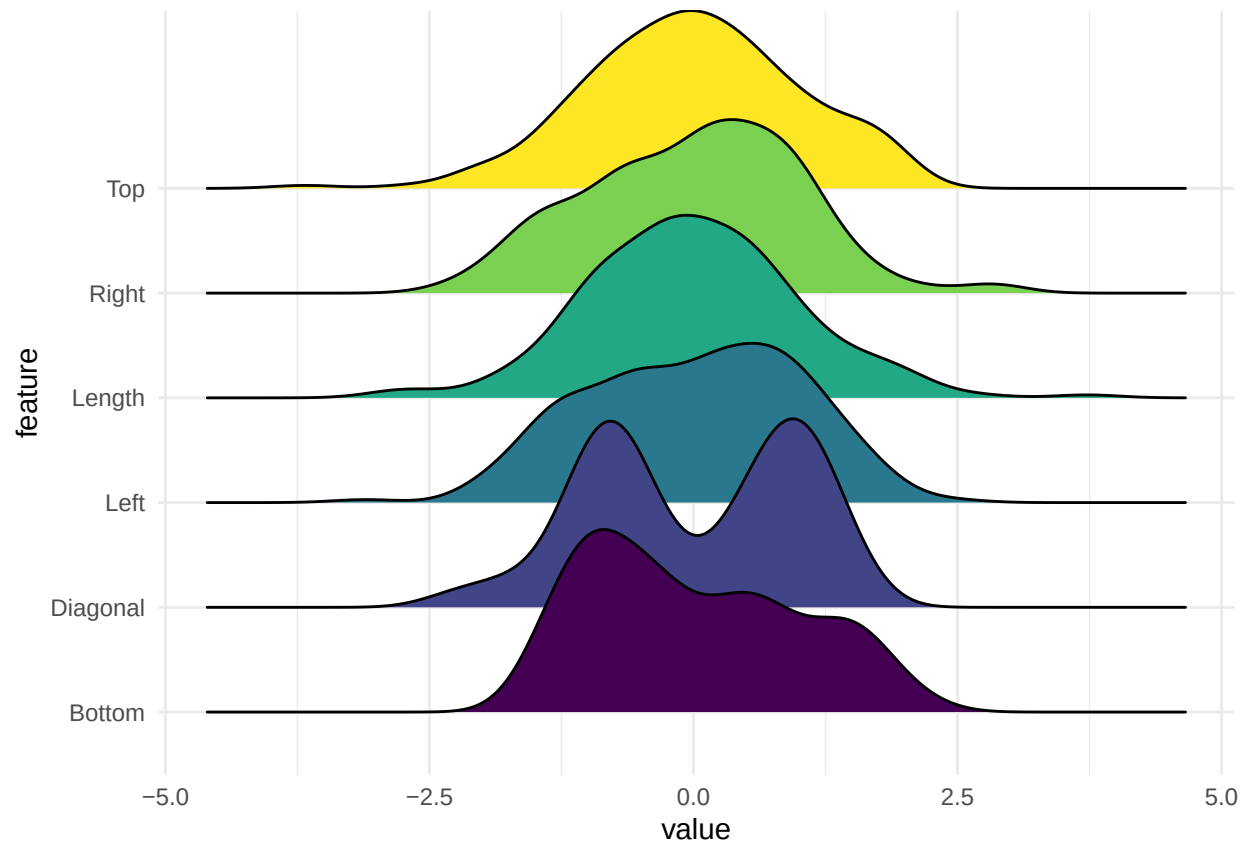
# density plot for all columns in the data set
df |> ggplot(aes(x = Length)) + geom_density() + theme_minimal() +
df |> ggplot(aes(x = Left)) + geom_density() + theme_minimal() +
df |> ggplot(aes(x = Right)) + geom_density() + theme_minimal() +
df |> ggplot(aes(x = Bottom)) + geom_density() + theme_minimal() +
df |> ggplot(aes(x = Top)) + geom_density() + theme_minimal() +
df |> ggplot(aes(x = Diagonal)) + geom_density() + theme_minimal()
```



The diagonal feature looks good. Look at the two bumps in its density plot

colorful alternative

```
df |> mutate(across(everything(), scale)) |>
  pivot_longer(everything(), names_to = "feature", values_to = "value") |>
  ggplot(aes(x = value, y = feature, fill = feature)) +
  geom_density_ridges() +
  scale_fill_viridis_d(guide = "none") +
  theme_minimal()
```



```
# 6 univariate density plots (Length, Left, Right, Bottom, Top, Diagonal) are
# compared to see which single feature shows the clearest separation into
# two groups (bimodal meaning 2 distinct bumps)
# Diagonal stands out with two clearly separated peaks in its density curve,
# while the others are mostly single-peaked (unimodal) or only weakly bimodal.
# Diagonal the best single candidate feature for clustering, since a two-humped
# density strongly suggests two underlying subpopulations (exactly what model-based
# clustering is designed to detect.
```

Univariate model-based clustering

Perform model-based clustering with 2 clusters on the **Diagonal** feature, assuming equal variance i.e. forcing both clusters to share the same variance

```
fit_E_2 <- Mclust(df$Diagonal, G = 2, modelNames = "E")

# means
fit_E_2$parameters$mean
```

```
##          1          2
## 139.4464 141.5221
```

```
# variance
fit_E_2$parameters$variance$sigma2
```

```
## [1] 0.244004
```

```
# Means (139.45 and 141.52), two distinct centre, consistent with the two
# humped density density seen earlier
# Variance (0.244), shared by both clusters, since equal clusters was assumed
```

Compute the model's BIC using its log-likelihood and compare it to the BIC stored in the model object.

```
# This model has 4 parameters (1 class probability pi, 2 means, and 1 variance)
# BIC computed
- 2 * fit_E_2$loglik + log(nrow(df)) * 4
```

```
## [1] 569.4667
```

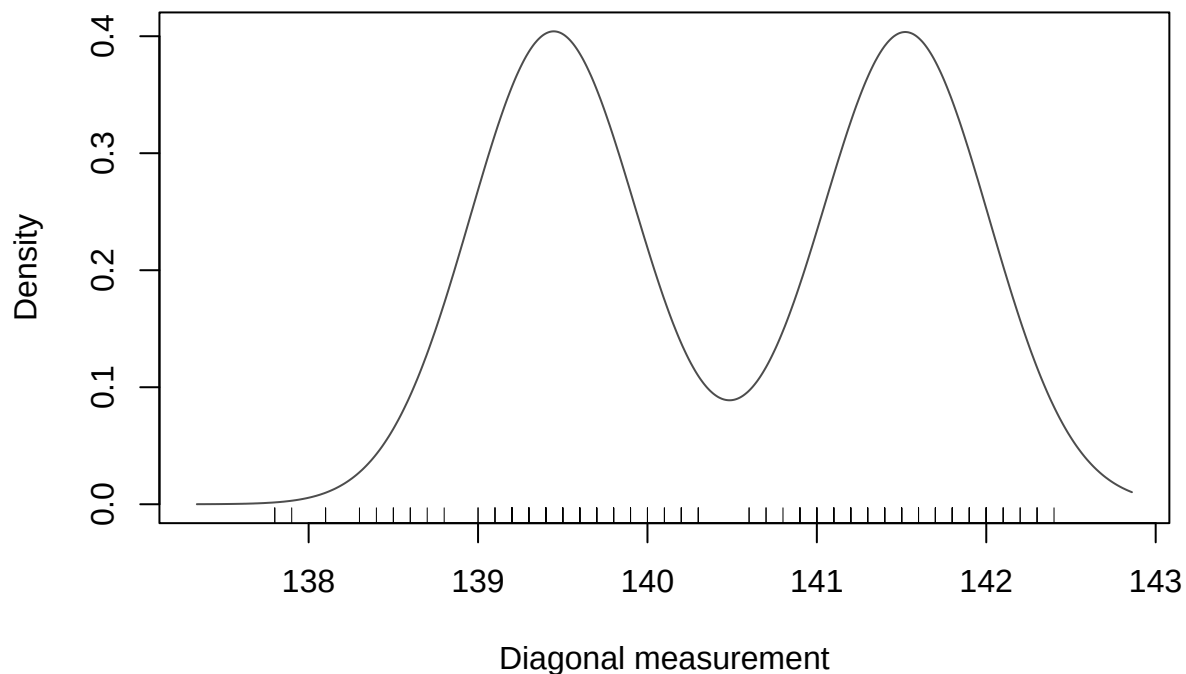
```
# BIC from mclust package
fit_E_2$bic
```

```
## [1] -569.4667
```

```
# BIC from the Mclust package is wrongly multiplied by -1

# manual BIC calculation using the formula (-2 × log-likelihood + log(n) × m,
# where m = 4 parameters i.e. 1 mixing proportion, 2 means, 1 shared variance)
# gives 569.47, matching fit_E_2$bic exactly (aside from a sign flip).
# Mclust stores BIC as the negative of the conventional definition,
# so fit_E_2$bic = -569.4667
# You need to negate it inorder to make comparissons
```

```
# plot the density using base R plots
plot(fit_E_2, "density", xlab = "Diagonal measurement")
# add the observations using rug marks
rug(df$Diagonal)
```



```
# shows two symmetric, equally-wide bell curves centered near 139.5 and 141.5,
# with rug marks below showing the actual raw data points.
# The two bumps have the same width because equal variance was forced.
```

Perform model-based clustering with 2 clusters on the Diagonal feature again, but now assume unequal variances.

```
# Let's set the seed for reproducibility
fit_V_2 <- Mclust(df$Diagonal, G = 2, modelNames = "V")
```

```
# mean
fit_V_2$parameters$mean
```

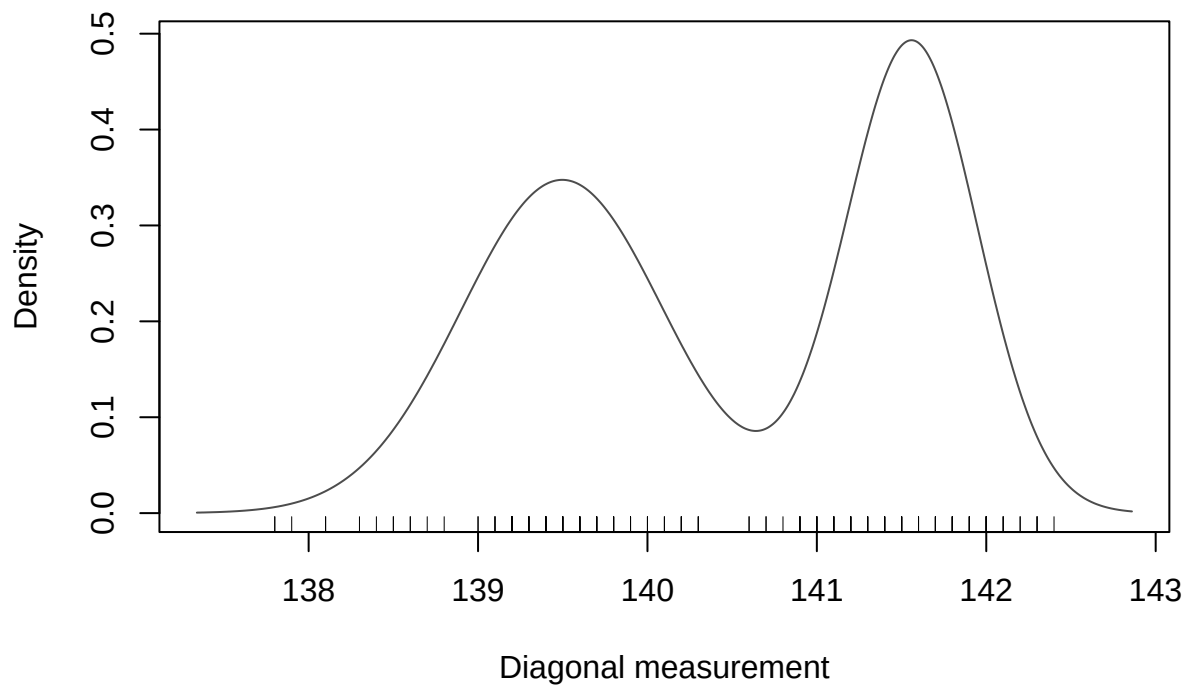
```
##           1           2
## 139.4973 141.5604
```

```
# variance
fit_V_2$parameters$variance$sigmasq
```

```
## [1] 0.3589844 0.1500838
```

```
# Means (139.50 and 141.56), nearly identical to the equal variance model
# Variance (0.359 and 0.150), now clearly different. The first cluster (lower Diagonal)
# is more spread out than the second
```

```
# plot the density using base R plots
plot(fit_V_2, "density", xlab = "Diagonal measurement")
# add the observations using rug marks
rug(df$Diagonal)
```



```
# The left bump is wider/shorter (more spread, less peaked) and the right bump
# is narrower/taller (more concentrated), directly reflecting the different variances.
```

```
# This model has 5 parameters: 1 class probability  $\pi$ , 2 means, and 2 variance
# BIC computed
-2*fit_V_2$loglik
```

```
## [1] 537.0199
```

```
# BIC from mclust package
fit_V_2$bic
```

```
## [1] -563.5115
```

Deviance Comparisson

```
# model with equal variance
-2*fit_E_2$loglik
```

```
## [1] 548.2735
```

```
# model with unequal variance
-2*fit_V_2$loglik
```

```
## [1] 537.0199
```

```
# Lower deviance means better fit to the data, so the V (unequal variance) model  
# fits better.  
# This makes sense, since it has an extra parameter giving it more flexibility  
# to match the data exactly.
```

BIC Comparisson

NB: Remember, the package shows negative BIC

```
# model with equal variance
-fit_E_2$bic
```

```
## [1] 569.4667
```

```
# model with unequal variance
-fit_V_2$bic
```

```
## [1] 563.5115
```

```
# The V model still wins on BIC (lower is better), but the gap between the two models  
# has narrowed compared to the deviance comparison.  
# This is exactly what BIC is designed to do. It penalizes the extra parameter in  
# the V model, so while V still fits better overall, its advantage is less  
# dramatic than raw deviance alone suggests. BIC is protecting against overfitting.
```

Multivariate model base clustering

We will now use all the available information in the data set to cluster the observations

```
# mlcust with all 6 features  
fit <- Mclust(df)  
  
summary(fit)
```

```
## -----
## Gaussian finite mixture model fitted by EM algorithm
## -----
##
## Mclust VVE (ellipsoidal, equal orientation) model with 3 components:
##
## log-likelihood  n df      BIC      ICL
##      -663.3814 200 53 -1607.574 -1607.71
##
## Clustering table:
##  1  2  3
## 18 98 84
```

```
# The optimal model is a VVE model with 3 components
```

```
# mean parameters
fit$parameters$mean
```

```
##           [,1]      [,2]      [,3]
## Length  215.023017 214.971360 214.78091
## Left    130.499684 129.929686 130.26435
## Right   130.304813 129.701143 130.17988
## Bottom   8.775842   8.301115  10.85719
## Top     11.173812  10.162379  11.10798
## Diagonal 138.731602 141.541673 139.62387
```

```
# 3 clusters times 6 variables = 18 mean parameters
```

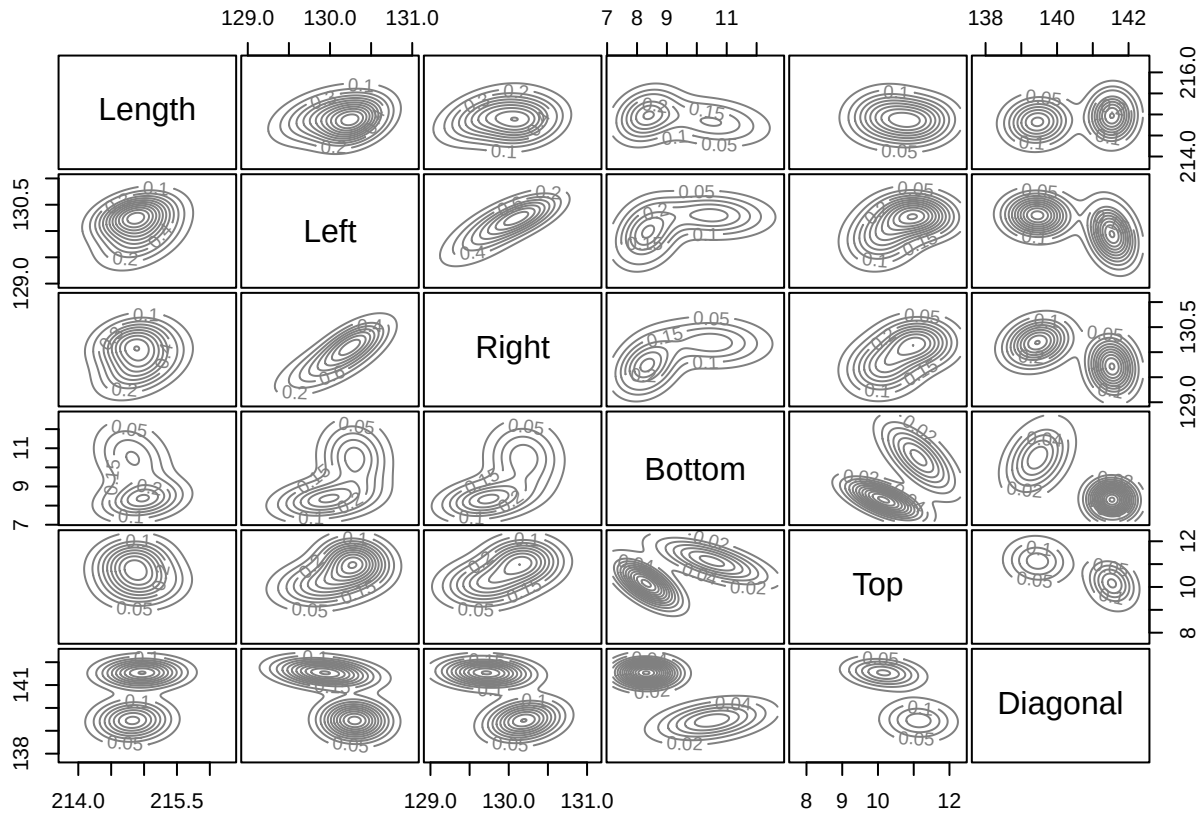
```
# Running Mclust(df) with no constraints lets it search across all covariance
# structures and 1-9 possible cluster counts, then pick whichever combination
# minimizes BIC. The winner: a VVE model with 3 components, meaning:
# Variable volume (cluster can have different sizes/spread)
# Variable shape (clusters can be different shapes, not just circular)
# Equal orientation (all clusters point the same direction in feature space)

# 3 clusters, with 18, 98, and 84 observations respectively (200 total)
# Multivariate clustering found 3 clusters, not 2, even though the data only has
# 2 true classes (genuine/counterfeit).
# The data may genuinely split into 3 identifiable subgroups, even though our
# external knowledge says there are only 2 real-world categories.
```

Forcing exactly 2 clusters (to match the known genuine/counterfeit split) with the most flexible covariance structure (VVV = variable volume, variable shape, variable orientation). The resulting matrix of bivariate density contour plots lets you inspect, for every pair of variables, how well-separated the two estimated clusters are.

```
# 2-component VVV model
VVV2 <- Mclust(df, 2, "VVV")

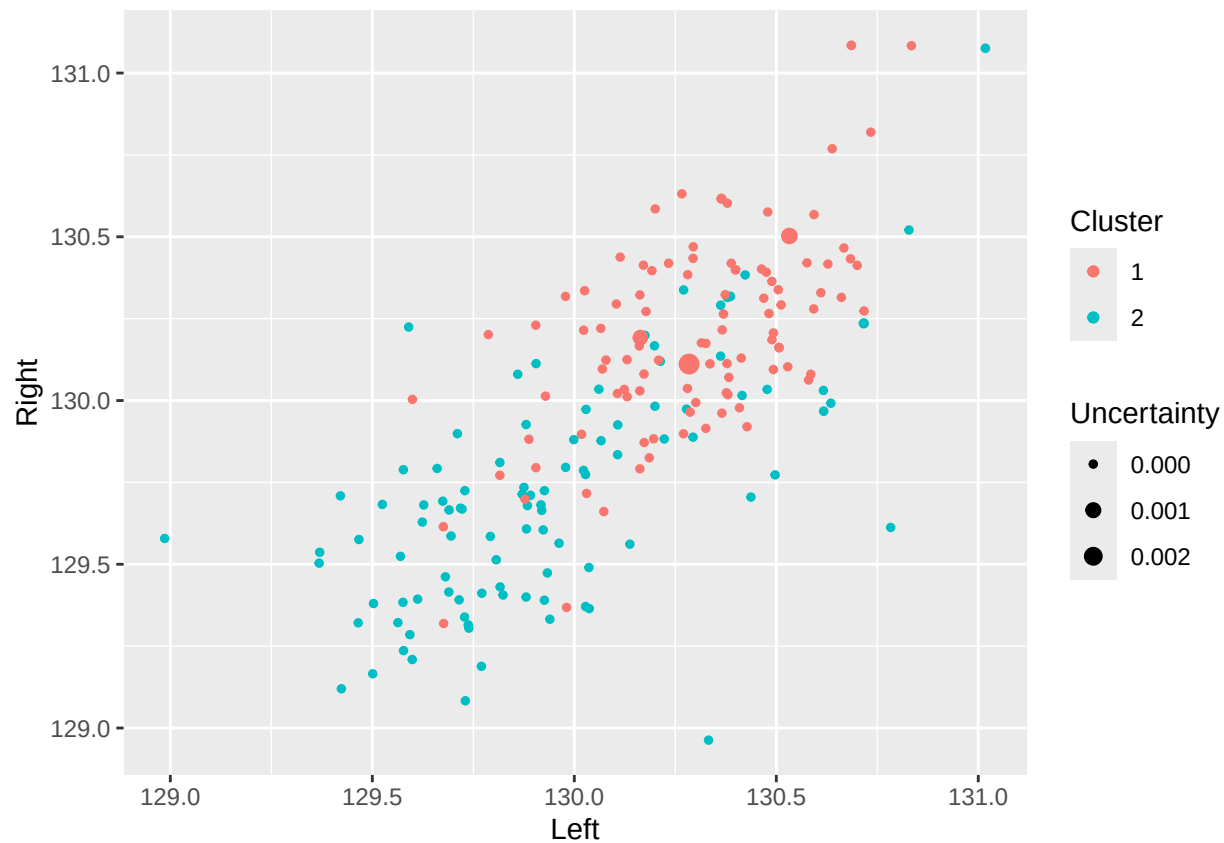
# matrix of bivariate contour (density) plots
plot(VVV2, "density")
```



*# Diagonal-top, and diagonal-bottom separate the classes very well
 # Diagonal-Top and Diagonal-Bottom pairs show the clearest separation
 # (two distinct, non-overlapping contour blobs), while most other pairs
 # (like Left-Right, matching the very first plot in the document) show heavily
 # overlapping contours, reinforcing that Diagonal is the single most discriminating
 # feature, and combining it with Top or Bottom gives especially clean separation.*

Plotting Left vs Right again, but now coloring points by the model's estimated cluster assignment (not the true Status labels) and sizing points by classification uncertainty.

```
df |> ggplot(aes(x = Left, y = Right, colour = as_factor(VV2$classification),
                 size = VV2$uncertainty)) +
  geom_point(position = position_jitter()) +
  scale_size(range = c(1, 3)) +
  labs(colour = "Cluster", size = "Uncertainty")
```



*# almost all points are small (low uncertainty) meaning the model is very confident
 # in most assignments, with only 3 points showing any meaningful uncertainty,
 # and notably, those uncertain points don't sit near the visual boundary between
 # the two colors in this particular Left/Right view.
 # only 3 points are slightly uncertain, and they are not around the border of the
 # classes in these two dimensions.
 # Apparently, the other dimensions give enough information
 # about those points near the border*